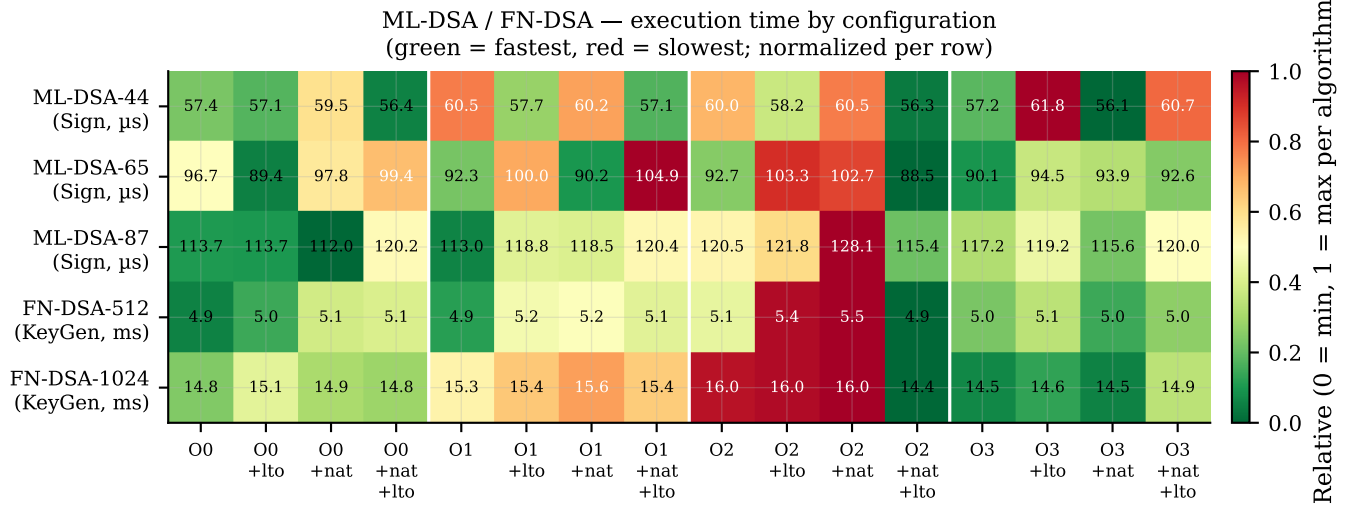
ML-DSA / FN-DSA — execution time by configuration
(green = fastest, red = slowest; normalized per row)
ML-DSA-44 (Sign, µs)
ML-DSA-65 (Sign, µs)
ML-DSA-87 (Sign, µs)
FN-DSA-512 (KeyGen, ms)
FN-DSA-1024 (KeyGen, ms)
57.4 57.1 59.5 56.4 60.5 57.7 60.2 57.1 60.0 58.2 60.5 56.3 57.2 61.8 56.1 60.7
96.7 89.4 97.8 99.4 92.3 100.0 90.2 104.9 92.7 103.3 102.7 88.5 90.1 94.5 93.9 92.6
113.7 113.7 112.0 120.2 113.0 118.8 118.5 120.4 120.5 121.8 128.1 115.4 117.2 119.2 115.6 120.0
4.9 5.0 5.1 5.1 4.9 5.2 5.2 5.1 5.1 5.4 5.5 4.9 5.0 5.1 5.0 5.0
14.8 15.1 14.9 14.8 15.3 15.4 15.6 15.4 16.0 16.0 16.0 14.4 14.5 14.6 14.5 14.9
O0
O0 +lto
O0 +nat
O0 +nat +lto
O1
O1 +lto
O1 +nat
O1 +nat +lto
O2
O2 +lto
O2 +nat
O2 +nat +lto
O3
O3 +lto
O3 +nat
O3 +nat +lto
Relative (0 = min, 1 = max per algorithm)
1.0
0.8
0.6
0.4
0.2
0.0